

Subhasish Bag, Ph.D.

Theoretical and Computational Plasma Physicist

JRG Post Doctoral Fellow

Asia Pacific Center for Theoretical Physics (APCTP), Pohang, South Korea.

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 LinkedIn  ResearchGate

 Plasma Simulation Group (PSG)  Plasma Physics Laboratory (PPL)

 Yoon Plasma Group (YPG)

Experience

- Nov, 2025- Sep, 2026  **Post Doc at Asia Pacific Center for Theoretical Physics (APCTP), Pohang, South Korea:** *Magnetohydrodynamics (MHD) Modeling of Magnetic Reconnection Thruster.*
- 2025 (Aug-Oct)  **Research Associate - I at Indian Institute of Technology Delhi (IIT Delhi):** *Modeling of finite size plasma.*
- 2025 (Jul-Aug)  **Early Doctoral Fellow at Indian Institute of Technology Delhi (IIT Delhi):** *Working on flowing magnetized plasma.*
- 2025 (Jan-Jul)  **Coordinator of the internal committee of National Energy Task Force at Indian Institute of Technology Delhi (IIT Delhi):** *Worked on developing a roadmap for the future energy sector: administrative and strategic planning activities.*

Education

- 2018 – 2025  **Ph.D., Indian Institute of Technology Delhi (IIT Delhi)** in Plasma Physics. Thesis title: *Fluid Modelling and Simulations of a finite size unmagnetised Plasma and a Flowing Magnetized Plasma.*
- 2021 – 2022  **Industry Sponsored Project (RESPOND Grant by LPSC-ISRO), Indian Institute of Technology Delhi (IIT Delhi)** in Plasma Physics (Not part of my thesis). Project title: *Classical Trajectory-Monte Carlo Simulations of the Plasma Dynamics in an Electrode-less Plasma Thruster.*
- 2017  **Summer Internship, Indian Institute of Science Education and Research (IISER) Kolkata.** Thesis title: *Generalized concurrence measure for hypergraph states using Lagrange's identity and wedge product.*
- 2016 – 2018  **M.Sc. (1st class), Indian Institute of Technology (Indian School of Mines), Dhanbad** in Physics- 1st class. Thesis title: *A Theoretical Investigation on Hydrogen Adsorption Capacity on Sc Decorated small Boron Cluster.*
- 2013 – 2016  **B.Sc. (1st class), Serampore College, University of Calcutta** in Physics (Hons.)

Research Interest

- Plasma Physics  **Theoretical and Computational: Fluid and Kinetic Simulation, Computational Fluid Dynamics for Plasma, Particle Trajectory and MC, MD and PIC simulations, Magnetized Plasma Modelling for Electric Propulsion and Material Processing Applications, Magnetic Nozzle, Magnetic Reconnection, Electrodeless Plasma Thruster, Plasma Wake Field Acceleration, Fusion Plasma Modeling, Modeling of Plasma Turbulence, Plasma Chemistry, AI/ML in Plasma Flow.**
- Quantum Computing  **Quantum Simulation, Quantum Technology, AI/ML.**

Completed Research Projects

Project-I	■ Flow modeling of magnetized expanding plasma of an Electron Cyclotron Resonance Thruster experiment and Material Processing Applications (Thesis work).
Project-II	■ Modeling of finite size plasma (Thesis work).
Project-III	■ Classical Trajectory - Monte Carlo Simulations of the Plasma dynamics in electrode-less plasma thruster (Industry sponsored Research work, not included in thesis).
Project-IV	■ Developing Fluid code based on Finite Volume Method (FVM) using Computational Fluid Dynamics approach (Not included in thesis).
Project-V	■ Studies of Quantum Graph and Hypergraph States for various quantum applications (based on research interest).
Ongoing Project	■ Magnetohydrodynamics (MHD) Modelling of Magnetic Reconnection Thruster.
Project Advisor	■ Physics-Informed Probabilistic Collapse Prediction Using Bouc-Wen Hysteretic Neural Networks Under Real Seismic Excitation. Selected as "Outstanding Research Teams" by APEC (Asia-Pacific Economic Cooperation) committee.

Research Skill

Computational Skill	■ Programming Languages, Software, Operating System Fortran, C, C++, Python, Matlab, Gaussian (DFT), Chemcraft, Ansys (Fluent), Comsol (AC-DC Module), EPOCH (PIC), MHD Flash Code, Codex, Parallel Programming, PyTorch, PINN, HPC, Microsoft Q#, IBM Qiskit, Linux, Windows, Mac.
Mentorship Skill	■ (1) MSc Physics student- Mrityunjoy Bauri (M.Sc., IIT Delhi): 2021-2022, Rohit (M.Sc., IIT Delhi): 2021-2022. Topic- "Reflection/transmission of laser-plasma solitons from localized plasma inhomogeneity". (2) B.Tech in Math and Computing student- Hitesh Yadav (B.Tech 2nd year, 2023, IIT Delhi). Topic- "Development of Electrostatic PIC code for Thruster Applications". (3) B.Tech in Engineering Physics student- Parth Verma (B.Tech 2nd year, Delhi Technological University (DTU), 2025). Topic- "Fundamental modes and breaking of oscillations for uniform plasma". (4) MSc Physics student- Vishal (M.Sc., IIT Delhi): 2025-2026, Vikas (M.Sc., IIT Delhi): 2025-2026. Topic- "Two temperature Model for Magnetized Flowing Plasma". (5) APEC (Asia-Pacific Economic Cooperation) Mentoring Research Activity (APEC MRA) under APEC Mentoring Center for the Gifted in Science (AMGS): Mentoring three international high school students for three months at South Korea (March-June, 2026). Theme- "Empowering Youth Innovation for Climate-Resilient and Sustainable APEC".
Teaching Assistantship	■ (1) PYP100: B. Tech. (UG-Lab Courses) 1st Year Electrical Lab: 2018-19 (Sem II), 2019-20 (Sem I,II) for six different electrical experiments (two at each semester). (2) PYL101: B. Tech. (UG-Theory Courses) 1st Year Electromagnetics and Quantum Mechanics : Tutorial and Coordination; 2020-21 (Sem II,I). (3) PYL657: Plasma Physics (PG-Theory Course): Tutorial and Coordination; 2021-22 (Sem I).

Research Skill (continued)

Organization Skill	(1) 2019 IIT Delhi Open House volunteer (outreach program). (2) Member of team in one day Interaction Seession on Plasma Physics on 11st June, 2022 at IIT Delhi (outreach program). (3) Member of organization team in Strong Plasma Radiation Interaction Group (SPRING-II) ; 9th-10th Mach, 2024 at IIT Delhi. (4) Student Secretary and Student General Secretary of a cultural and social samity named Basanta Utsav Samity of IIT Delhi. (5) Student Head Coordinator of National Workshop on Future Ready Energy Education: Opportunities and Challenges held at IIT Delhi on 7th July, 2025. (6) Member of organization team of Strong Plasma Radiation Interaction Group – III (SPRInG-III) held at IIT Delhi on 7-8 March 2026.
Membership	(1) Student member of Astro-Club of Department of Physics at IIT(ISM) Dhanbad. (2) Lifetime Membership of Plasma Science Society of India (PSSI)- LM-1603. (3) Member of Electric Rocket Propulsion Society(ERPS), the international society of electric propulsion. (4) Member of American Physical Society Division of Plasma Physics (APS DPP)- 62133123. (5) Member of Association of Asia Pacific Physical Societies Division of Plasma Physics (AAPPS DPP)-2528. (6) Member of American Institute of Aeronautics and Astronautics (AIAA)- 1405726 . (7) Member of International Low Temperature Plasma Community (ILTPC). (8) Reviewer of Physics of Plasmas (PoP) journal: journal of AIP (American Institute of Physics) Publishing; Journal of the Korean Physical Society:of Springer.
Communication Skills	English, Bengali, Hindi- Fluent (speaking, reading, writing, listening); Learning French and Korean.
Visit	Two days (29-30 September, 2022) lab visit and presented the research work about Electron Cyclotron Resonance (ECR) Thruster Modeling in the Group of Prof Kazunori Takahashi in the Department of Electrical Engineering, Tohoku University, Sendai, Japan.

Non-academic Interest

Extracurricular Activities	Football, Swimming, Running, Reading, Indian Classical Vocal Music, Playing Musical Instruments (Harmonium, Tabla, Guitar), Acting, Event Management, Traveling and Exploring New Places, Ancient History, and Learning New Languages.
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Completed Research Projects

- Project-I  **Flow modeling of magnetized expanding plasma of an Electron Cyclotron Resonance Thruster experiment and Material Processing Applications.**
(1) Relevance: Magnetized plasma expansion is crucial for electric spacecraft propulsion, addressing fuel inefficiency and low-thrust demands for orbit adjustments in deep-space missions. We have issues like electrode erosion and contamination that challenge conventional propulsion methods. Advanced systems with magnetic nozzles (MN) offer advantages by avoiding electrode-related problems with converting electrons' thermal energy into ions' kinetic energy for thrust.
(2) Model: A new model has been developed in the magnetic coordinate system and implemented a new algorithm to solve the plasma flow equations..
(3) Result: We implement our model for the small volume plasma experiment of the Compact ECR Plasma Source (CEPS) at IIT Delhi.
(4) Collaborator: Prof. Ashish Ganguli (PI), Prof. Ramesh Narayanan, Prof Debaprasad Sahu; Plasma Physics Lab, Department of Energy Science and Engineering (DESE), .
- Project-II  **Modeling of finite size plasma**
(1) Description- Recent studies explore small plasma formations created when liquid droplets are struck by lasers. These investigations aim to understand the linear and non-linear dynamics of plasma droplets, including their expansion behavior. Notably, the finite size of the plasma introduces edge inhomogeneities, leading to distinct normal mode behaviors compared to infinite plasma media.
(2) Project: In this project, we model finite-size plasma to investigate plasma dynamics and address fundamental problems using fluid simulations.
(3) Collaborator: Prof. Amita Das; Department of Physics, IIT Delhi.
- Project-III  **Classical Trajectory - Monte Carlo Simulations of the Plasma dynamics in electrode-less plasma thruster.**
(1) Description- We attempted to simulate plasma dynamics in an electron cyclotron resonance (ECR) thruster using a classical trajectory-based code.
(2) Result- Our results show that assuming electrons follow n/B scaling and a Boltzmann distribution, while evolving ions with the trajectory code, yields thrust estimates consistent with experimental data. However, incorporating self-consistent dynamics of electrons and ions proved computationally inefficient due to the high ion-to-electron mass and temperature ratios. We conclude that these classical trajectory methods are unsuitable for self-consistent simulations of plasma expansion in ECR thrusters, and recommend alternative approaches like full particle-in-cell (PIC) simulations or hybrid models with fluid electrons and PIC ions.
(3) Collaborator: Dr. Asish James; Liquid Propulsion Systems Centre (LPSC)/Indian Space Research Organization (ISRO).
- Project-IV  **Developing Fluid code based on Finite Volume Method (FVM) using Computational Fluid Dynamics approach.**
Description- Based on the finite volume method, the wave code has been developed to study the plasma dynamics. We explore basic plasma nonlinear phenomena using this code.

Completed Research Projects (continued)

- Project-V  **Studies of Quantum Graph and Hypergraph States for various quantum applications.**
(1)**Description-** Graph states, characterized by high entanglement can be mathematically associated with graphs, where vertices and edges correspond to quantum spin systems and Ising interactions, respectively. These states are versatile and can be mapped onto various regular 2D lattices. Hypergraph states, an extension of graph states with more than two connected vertices by a single edge, offer extensive utility in multiple domains. These states are not only essential in quantum error correction and quantum blockchain but have also found applications in neural networks. We study these states for various quantum applications.
(2)**Result-** (i) We have classified these states based on their geometry. (ii) Study of metrological usefulness of Graph and Hypergraph states. (iii) Study of Absolutely Maximally Entangled (AME) Graph and Hypergraph States.
(3) **Collaborator:** Prof. Prasanta K. Panigrahi; Center for Quantum Science and Technology, Siksha 'O' Anusandhan University, and Department of Physical Sciences, Indian Institute of Science Education and Research Kolkata, India; Dr. Ramita Sarkar; Institute of Physics Bhubaneswar (IOP), India.
- Ongoing Project  **Magnetohydrodynamics (MHD) Modelling of Magnetic Reconnection Thruster.**
(1) **Description-** A new concept for magnetic reconnection has been implemented which will be the more effective for the designing of magnetic reconnection thruster.
(2) **Group:** Prof. Young Dae Yoon, Junior Research Group Leader at Asia Pacific Center for Theoretical Physics; Affiliated Professor at Pohang University of Science and Technology (POSTECH), Pohang, Republic of Korea.

Research Publications (Published, Submitted and Under Preparation)

Published Articles:

- 1 Bag, Subhasish, V. Saxena, and A. Das, "Fluid simulations for a finite size plasma," *Physics of Plasmas* (Published; IF: 2.2), vol. 31, no. 7, p. 072 105, 2024.
- 2 Bag, Subhasish and V. Saxena, "Nonlinear evolution of a one dimensional cold non-relativistic electron-ion plasma with an arbitrary initial density profile: A phase mixing perspective," *Physica Scripta* (Published; IF: 2.6), vol. 100, no. 3, p. 035 616, 2025.
- 3 Bag, Subhasish, R. Sarkar, and P. K. Panigrahi, "A concurrence-based criterion for equal maximally entangled and absolutely maximally entangled states," *Quantum Information Processing* (Published; IF: 2.2), vol. 25, no. 2, p. 60, 2026.
- 4 R. Sarkar, S. Banerjee, Bag, Subhasish, and P. K. Panigrahi, "Geometry of distributive multiparty entanglement in 4- qubit hypergraph states," *IET Quantum Communication* (Published; IF: 2.8), vol. 3, no. 1, pp. 72–84, 2022.

Filed Patent:

- 1 Bag, Subhasish, A. Ganguli, V. Saxena, D. Sahu, and R. Narayanan, *Analysis of flowing plasma in a magnetic field for plasma based processing and thruster applications*, (FT/IDF/3/2026/61).

Manuscripts under Submission:

- 1 Bag, Subhasish, A. Ganguli, V. Saxena, D. Sahu, and R. Narayanan, *Modelling of flowing plasma in the magnetic field of the small volume plasma system experiment*, 2026.
- 2 P. Singh, R. Narayanan, A. Ganguli, D. Sahu, and Bag, Subhasish, *Experimental and theoretical study of flow dynamics of expanding hydrogen plasma from an ecr plasma source*, 2026.

- 3 Bag, Subhasish, R. Sarkar, and P. K. Panigrahi, *Investigation of three qutrit pure absolutely maximally entangled special states*, 2026.
- 4 Bag, Subhasish, R. Sarkar, and P. K. Panigrahi, *Achieving heisenberg limit in the phase measurement through three-qubit graph states*, 2022.

Manuscripts under Preparation:

- 1 Bag, Subhasish, A. Ganguli, V. Saxena, D. Sahu, and R. Narayanan, *Algorithm for the construction of magnetic grid of small volume plasma system experiment*, 2026.
- 2 Bag, Subhasish et al., *Review on phase mixing induced wave breaking*, 2026.

Talk

- 1 Title- "Simulation of Expanding Plasma for Electron Cyclotron Resonance Thruster (ECRT)" At 4th Departmental Symposium on **Advanced in Physics** (1-3 December, 2020) by Department of Physics, IIT Delhi.
- 2 Title- "Dance of Charged Particles in a Plasma Zoo" Organized and Presented in one day **Interaction Session on Plasma Physics** on 11th June, 2022 at IIT Delhi.
- 3 Title- "One Dimensional Studies of Electrostatic Modes in Microplasma" At the **7th Asia-Pacific Conference on Plasma Physics (AAPS-DPP 2023)**; November 12-17, 2023, Port Messe Nagoya, Japan.
- 4 Title- "Exploring One-Dimensional Electrostatic Modes in Finite-Size Plasma" At **16th International Conference on Plasma Science and Applications (ICPSA 2023)** from 12th to 14th December, 2023 (online).
- 5 Title- "A Two-Fluid Approach to Study the Dynamics of Finite Size Plasmas" at **66th Annual Meeting of the APS Division of Plasma Physics (APS DPP 2024)**, October 7-11, 2024 in Atlanta, Georgia, USA.
- 6 Invited Talk- "Review of Quantum Computing Approaches in Plasma Physics Research." At the **Quantum Science and Quantum Technology (QSQT) 2025**, July 8-16, 2025, organized by Centre for Quantum Science and Technology (CQST), India (virtually).
- 7 Flash Talk- "Modeling of Flowing Magnetized plasma for Small Volume Plasma System." at **PPPL Graduate Summer School (GSS-2025) on Plasmas for Microelectronics and Quantum Information Science** ; July 28 - August 1, 2025 by PPPL (Princeton Plasma Physics Laboratory) virtually.
- 8 Invited Talk- "Modeling of flowing magnetized plasma for small volume plasma system" at the **9th Asia-Pacific Conference on Plasma Physics (AAPS-DPP 2025)**, September 21-26, 2025, in Fukuoka International Congress Center, Japan.
- 9 Title- "Investigation of the dynamics of finite size plasma." at **67th Annual Meeting of the APS Division of Plasma Physics (APS DPP 2025)**, November 17-21, 2025 in Long Beach, California, USA.
- 10 Talk- "A Criterion for Investigating Absolutely Maximally Entangled Pure States." At the **Quantum Information and Emerging Quantum Technologies (QIET) 2026**, 27-31 January 2026, organized by National Institute of Technology Rourkela (NIT Rourkela), India (virtually).
- 11 Invited Talk- "Emerging Plasma Technologies in Space Propulsion." At the **Strong Plasma Radiation Interaction Group – III (SPRInG-III)**, 7-8 March 2026, organized by Indian Institute of Technology Delhi (IIT Delhi), India.
- 12 Talk- "An Asymmetric Current-Driven Magnetic Reconnection Thruster" At the **2nd Plasma Astrophysics School and Workshop (PASW26)**, May 12-15, 2026 at Busan, South Korea.

Poster

- 1 Title- "Simulating Plasma Dynamics in a Magnetic Nozzle" At **36th Symposium on Plasma Science and Technology (Plasma-2021)**; 13-15 December, 2021 organized by BITS Mesra in association with PSSI.
- 2 Title- "Numerical Simulation for the Investigation of Expanding Magnetized Plasma in an ECR Thruster Experiment" At **27th International Conference on Numerical Simulation of Plasmas (ICNSP) 2022** (30th August to 2nd September) virtually.
- 3 Title- "Numerical Simulations of the Plasma Dynamics in an ECR Thruster Experiment" At **11th International Conference on Reactive Plasmas / 2022 Gaseous Electronics Conference (ICRP-11 / GEC 2022)** on October 3-7, 2022 Sendai International Center, Sendai, Japan.
- 4 Title- "Trajectory Simulations of an Expanding Plasma generated from an ECR Plasma Source" At **25th International Workshop on Electron Cyclotron Resonance Ion sources (ECRIS-2022)** virtually on 12-14 October 2022.
- 5 Title- "A Two-Fluid Approach To Study The Thrust Performance Of An ECR Thruster Experiment" At **37th National Symposium on Plasma Science and Technology (PLASMA 2022)** 12th to 14th December 2022 at IIT Jodhpur in association with Plasma Science Society of India (PSSI).
- 6 Title- "Thrust Measurement of An Electron Cyclotron Resonance Thruster Experiment Using A Two-Fluid Approach" At **3rd Conference on Plasma Simulation (CPS-2022)** 13-15 July 2023 Raman Science Center (IIA), Leh, Ladakh.
- 7 Title- "One Dimensional Studies of Electrostatic Modes in Microplasma" at the **7th Asia -Pacific Conference on Plasma Physics (AAPPS DPP 2023)**; November 12-17, 2023, Port Messe Nagoya, Japan.
- 8 Title- "One dimensional studies of electrostatic modes in microplasma" at **Strong Plasma Radiation Interaction Group (SPRING-II)** ; 9th-10th March, 2024 at IIT Delhi.
- 9 Title- "One Dimensional Studies Of Finite Size Plasma" at **1st Global Forum and International Workshop in Hybrid Mode on Industrial Plasma Processes and Diagnostics 2024 (IPPD 2024)**; May 9-10, 2024 at IIT Delhi.
- 10 Title- "Modeling of Plasma Thruster for Space Propulsion" at **Open House 2024**; 9th November, 2024 at IIT Delhi (**outreach program**).
- 11 Title- "Exploring the Dynamics of Finite Size Plasmas through Fluid Simulation" at **4th Conference on Plasma Simulation (CPS-2024)**; November 11-13, 2024 at Institute of Geomagnetism (IIG), Mumbai.
- 12 Title- "Modeling of Magnetized Flowing plasma for Small Volume Plasma Experiment." at **14th ITER International School (IIS-2025)**; June 30th to July 4th 2025 at Aix-Marseille University, Aix-en-Provence, France.

Other Participation

- 1 **Science exhibition**-demonstrate a model- 'rain alarm' made with 6 college students (2013 november-22 and 23, 2013) serampore college, hooghly.
- 2 Completed National Programme on Technology Enhanced Learning (NPTEL) 8 week course (July to September, 2017) on **Quantum Computation and Information** by IIT Bombay.
- 3 Attended the **33rd National Symposium on Plasma Science and Technology (PLASMA- 2018)** December 4- 7, 2018 organized by Department of Physics and Astrophysics, University of Delhi, Delhi In Association with Plasma Science Society of India (PSSI).

- 4 Attended two-days school on **ECR Ion Source Technology - Opportunities and Challenges**, at IUAC during 5-6 November 2020 (in online mode).
- 5 Attended **Quantum Information and Quantum Technology - 2022 (QIQT -2022)** school organized by IISER Kolkata 1st June to 20th June virtually.
- 6 Attended workshop on **Recent Developments in Plasma based Ion Sources for Accelerators and Associated Physics Programmes** during 16 -17 August, 2022, at IUAC New Delhi.
- 7 The **14th International School for Space Simulations: ISSS-14** (online), September 12-16th, 2022 by Kobe University.
- 8 Participated in AAPPs-DPP2022 as on-line Conference **6th Asia Pacific Conference on Plasma Physics** 9-14 October 2022 Division of Plasma Physics, Association of Asia-Pacific Physical Societies recommended by a participant.
- 9 Seminar: **Plasma for Space and Aerospace Application PSAA-2022** seminar on 24th and 25th of November at FCIPT, IPR-Gandhinagar (virtually).
- 10 School: **Introduction to Fusion Energy and Plasma Physics Course** June 5th-15th 2023 by PPPL (Princeton Plasma Physics Laboratory) virtually.
- 11 School: **2025 PPPL Graduate Summer School (GSS)** on Plasmas for Microelectronics and Quantum Information Science July 28 - August 1, 2025 by PPPL (Princeton Plasma Physics Laboratory) virtually.

Awards and Achievements

- 1 Awarded **Dr B.R.Ambedkar Medha Puraskar 2011** by the Government of West Bengal, in recognition for the meritorious performance in Madhyamik Pariksha.
- 2 Recipient of National Merit Scholarship- **Scholarship for Higher Education (SHE) of DST-INSPIRE**, Reg. No-201400009766 by Department of Science and Technology (DST)-New Delhi from 2013-2018 for five years.
- 3 Received of **Mentorship Grant** by Department of Science and Technology (DST)-New Delhi for the summer internship from IISERK in the field of quantum information and computing for MSc mandatory external project, 2017.
- 4 Grant- **GEC Student Travel Grant** from American Physical Society (APS) for attending and presenting research work at 75th Annual Gaseous Electronics Conference (GEC-2022) in Sendai, Japan.
- 5 Grant- **RSTA (Research Scholar Travel Grant)** from IIT Delhi for attending and presenting research work at 75th Annual Gaseous Electronics Conference (GEC-2022) in Sendai, Japan.
- 6 Grant- **Partial Conference Support** from AAPPs-DPP-2023 for attending and presenting research work at AAPPs-DPP 2023 conference (7th Asia-Pacific Conference on Plasma Physics; November 12-17, 2023, Port Messe Nagoya, Japan).
- 7 Grant- **APS Student Travel/Participation Grant** from American Physical Society (APS) for attending and presenting research work at 66th Annual Meeting of the APS Division of Plasma Physics (APS DPP 2024) in Atlanta, Georgia, USA.
- 8 Grant- **CSIR Student Travel Grant** from Council of Scientific and Industrial Research (CSIR) for attending and presenting research work at 66th Annual Meeting of the APS Division of Plasma Physics (APS DPP 2024) in Atlanta, Georgia, USA.
- 9 Grant- **ANRF DST Student Travel Grant** from Anusandhan National Research Foundation (ANRF) by the Department of Science and Technology (DST), New Delhi for attending and presenting research work at 14th ITER International School (IIS-2025) in Aix-Marseille University, Aix-en-Provence, France.

- 10 Grant- **APS Student Travel/Participation Grant** from American Physical Society (APS) for attending and presenting research work at 66th Annual Meeting of the APS Division of Plasma Physics (APS DPP 2025) in Long Beach, California, USA.

References

- 1 (Postdoc Leader) Name - **Prof. Young Dae Yoon**. Designation - Incoming Assistant Professor at UCLA; Junior Research Group Leader at Asia Pacific Center for Theoretical Physics; Affiliated Professor at Pohang University of Science and Technology (POSTECH). Email. Id – youngdae.yoon@apctp.org. Link: <https://yoonplasmagroup.com/> (For reference letter).
- 2 (PhD Supervisor) Name - **Prof. Vikrant Saxena**. Designation - Associate Professor. Institute – Indian Institute of Technology Delhi [IIT Delhi]. Email. Id – vsaxena@physics.iitd.ac.in, Ph: 01126591392. Department – Physics. Link: <https://web.iitd.ac.in/~vsaxena/> (For reference letter).
- 3 (PhD Collaborator) Name- **Prof Ashish Ganguli** . Designation – Professor(retired). Institute – Indian Institute of Technology Delhi [IIT Delhi]. Email. Id – ashishganguli@gmail.com, Department – Energy Science and Engineering. Link: <https://dese.iitd.ac.in/faculty-profile/34>.
- 4 (PhD Collaborator) Name - **Prof Amita Das**. Designation – Professor. Institute – Indian Institute of Technology Delhi [IIT Delhi]. Email. Id – amita@physics.iitd.ac.in, Department – Physics. Link: <https://web.iitd.ac.in/~amita/>.
- 5 (PhD Collaborator) Name- **Prof Ramesh Narayanan** . Designation – Professor. Institute – Indian Institute of Technology Delhi [IIT Delhi]. Email. Id – rams@dese.iitd.ac.in, Department – Energy Science and Engineering. Link: <https://ces.iitd.ac.in/faculty-profile/11>.
- 6 (PhD Collaborator and Member of my research committee) Name- **Prof Debaprasad Sahu**. Designation – Associate Professor. Institute – Indian Institute of Technology Delhi [IIT Delhi]. Email. Id – dpsahu@dese.iitd.ac.in, Department – Energy Science and Engineering. Link: <https://dese.iitd.ac.in/faculty-profile/16>.
- 7 (Summer Project Guide and Collaborator) Name – **Prof. Prasanta K. Panigrahi**. Designation - Professor (Director), Institute – Center for Quantum Science and Technology, Siksha 'O' Anusandhan University, and India Institute – Indian Institute of Science Education and Research Kolkata (IISERK). Email. Id – pprasanta@iiserk.ac.in; Department – Physical Sciences. Link: <https://sites.google.com/view/cqst-soadu/people?authuser=0>.